



Gesture-to-Voice over 5G: Real-Time Assistive Communication Platform

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Abstract

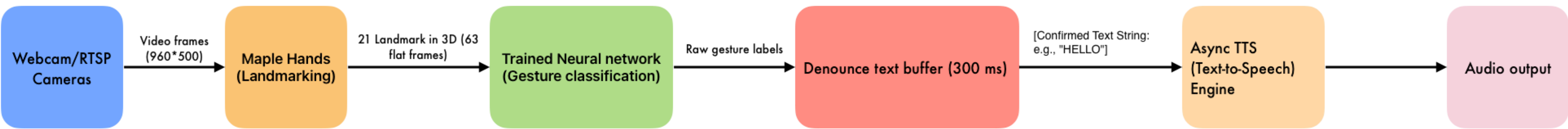
Design, Implementation, and evaluation of a Gesture to Voice software solution to enable accessibility for differently abled people.

Main Results

- Built a Python app using MediaPipe to extract 21 hand landmarks and an MLP classifier.
- Integrated a debounced text buffer with asynchronous, multi-backend TTS for low-latency, non-blocking spoken output.
- Developed an OpenCV interface with rule-based control gestures for clearing, spacing, and speaking sentences.

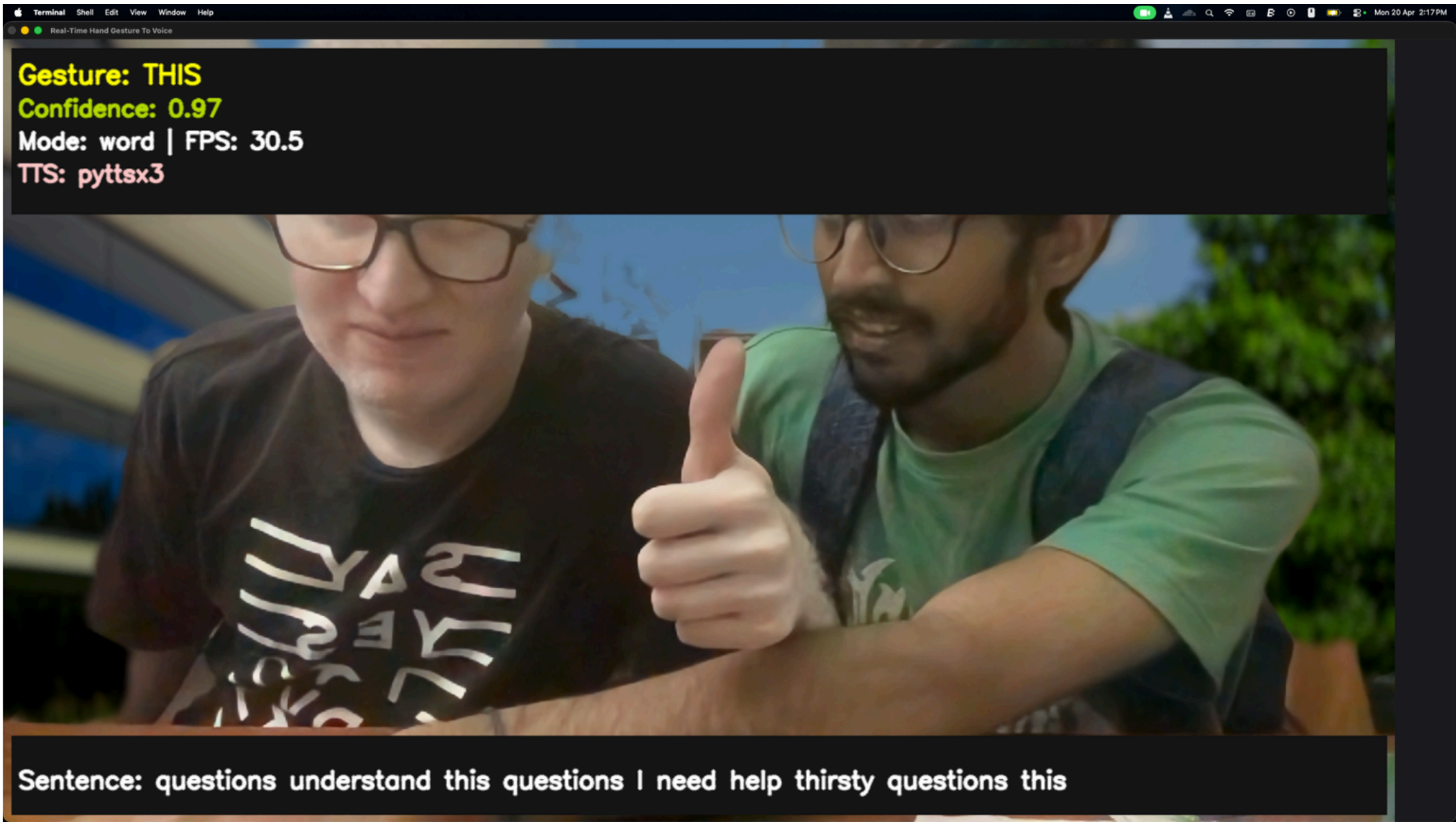
Technical Working

The software processes video frames through a pipeline that extracts 63 normalized landmark coordinates from a single hand using MediaPipe. These features are fed into a scikit-learn MLP classifier—or a rule-based fallback—which predicts gestures and updates a debounced text buffer. To ensure low-latency performance, the Text-to-Speech engine runs asynchronously in a separate worker thread, preventing audio synthesis from stalling the vision loop. While the app can use local webcams, it is designed for wireless and remote flexibility by accepting RTSP stream URLs as a camera source. This allows the software to ingest live video feeds from smartphones or IP cameras connected over a 5G network or local Wi-Fi, enabling the inference engine to run on a centralized edge device or laptop while the user remains mobile with a portable camera.



User Interface

To use the software, position your hand in front of the webcam to allow MediaPipe to extract 21 landmarks for real-time classification. The system operates on a three-second inference interval where the UI displays a live feed and blinks a green border to signal when a frame is being captured and processed. This design is fully asynchronous, using a dedicated worker thread for text-to-speech so that the vision loop remains low-latency and never stalls during audio output. Users manage the sentence buffer via an OpenCV overlay using keyboard shortcuts: press 's' to speak the current sentence, 'c' to clear the buffer, or 'm' to toggle between word and character modes.



MLP Vocabulary Reference		
Checked-in model: A small phrase-only MLP (5 classes). See models/labels.json and data/word_map_mini.json.		
Label	Spoken Output	Demo Image
GOOD_MORNING	good morning	
HAPPY_BIRTHDAY	happy birthday	
HOW_ARE_YOU	how are you	
I_NEED_HELP	I need help	
PLEASE	please	